

FINAL PROJECT REPORT

Strategic Employee Attrition Intelligence Dashboard

1. Introduction

1.1. Project Overview

The Strategic Employee Attrition Intelligence Dashboard is a Power BI-based workforce analytics project developed to understand employee attrition and identify patterns related to retention, performance, career growth, workload, work-life balance, and work arrangements. The project converts employee-level data into interactive dashboards and reports that can support HR and leadership decision-making.

The dashboards present key indicators such as total employees, employees who left, attrition rate, average burnout score, work-life balance score, employee engagement, manager support, performance rating, work experience, and overtime. Interactive filters are provided for education level, work mode, gender, and marital status.

The analysis focuses on attrition across salary-growth bands, education backgrounds, work models, tenure, career levels, overtime bands, performance levels, training investment, career stages, marital status, departments, and job roles.

1.2. Objectives

1. Quantify overall and segment-level attrition rates across department, job role, and work mode.
2. Identify the strongest factors associated with employee attrition, including burnout, overtime, compensation growth, tenure, and performance.
3. Build an interactive Power BI dashboard and report for HR and leadership decision-making.
4. Provide AI-assisted insights to support root-cause analysis.
5. Deliver actionable recommendations to reduce employee attrition and improve workforce wellbeing.

2. Project Initialization and Planning Phase

2.1. Define Problem Statement

The organization needs to understand why 895 of its 5,000 employees (17.9%) have left, and which factors—compensation, workload, engagement, career progression, or work arrangement—most strongly predict attrition. Without this understanding, HR cannot prioritize retention spending or design targeted interventions.

The project therefore focuses on transforming employee workforce data into measurable attrition indicators and interactive visual analysis so that patterns can be examined across relevant employee and workplace segments.

2.2. Project Proposal (Proposed Solution)

The proposed solution is an interactive Power BI analytics solution consisting of dashboards and reports for employee retention, performance, career growth, and workforce attrition analysis. The solution combines KPI cards, charts, slicers, and analytical visuals to allow users to explore attrition patterns from multiple dimensions.

The solution helps HR users move from a high-level attrition measurement to segment-level analysis. The dashboard presents workforce KPIs first and then provides visual breakdowns by salary growth,

education, work model, tenure, career level, overtime, performance, training investment, career stage, marital status, department, and job role.

AI-assisted insights are included as part of the analytical approach to support root-cause exploration. The outputs are intended to help identify areas where retention initiatives and workforce wellbeing actions can be considered.

2.3. Initial Project Planning

The project was completed individually. The planned project period was from the 10th to the 22nd. Since the work was carried out by one person, the sprint activities represent individual development phases rather than separate team members.

Sprint / Phase	Planned Activities	Output
Sprint 1	Project definition, problem statement, objectives, data identification and initial planning.	Project scope and initial plan
Sprint 2	Data collection, data quality review, exploration and preprocessing.	Prepared employee attrition dataset
Sprint 3	Business-question framing and Power BI visualization development.	Interactive charts, KPIs and slicers
Sprint 4	Dashboard/report design, insights, testing and final documentation.	Final dashboards and documentation

3. Data Collection and Preprocessing Phase

3.1. Data Collection Plan and Raw Data Sources Identified

The project uses an employee attrition dataset containing employee-level workforce information. The analysis uses attributes related to employee status, department, job role, work mode, education, compensation growth, workload/overtime, performance, tenure, career level, training investment, manager support, engagement, burnout, work-life balance, and demographic segments.

The raw data was used as the source for Power BI analysis. The data collection plan focused on ensuring that the fields required to answer the defined business questions were available before visualization development.

3.2. Data Quality Report

Data quality was reviewed before dashboard development to ensure that fields required for the analysis could be used consistently. The main quality checks focused on missing values, consistency of categorical values, numerical fields, attrition/status fields, and suitability of fields for aggregation and filtering.

Data Quality Area	Issue / Check	Resolution / Action
Missing values	Check employee attributes used in KPIs and visuals for blank values.	Review blanks before aggregation and handle records appropriately.
Categorical consistency	Department, work mode, education, gender, marital status and other categories must be consistent.	Standardize category values before visualization.
Numerical fields	Scores, experience, overtime and other numeric fields need suitable data types.	Set suitable numeric data types and aggregations.
Attrition status	Employee status is required to distinguish employees who left from employees who remained.	Use a consistent attrition indicator for calculations.
Duplicate records	Employee-level analysis requires unique employee records.	Check employee identifiers for duplicates.

3.3. Data Exploration and Preprocessing

The dataset was explored to understand the available employee attributes and their suitability for the project objectives. Fields were categorized into employee demographics, employment characteristics, performance and career variables, workload variables, wellbeing variables, and attrition indicators.

The preprocessing stage included checking data types, preparing categorical fields for slicers and grouping, preparing numerical fields for aggregation, and creating the calculations required for KPI cards and attrition-rate analysis. The prepared data was then loaded into Power BI for visualization and interactive analysis.

4. Data Visualization

4.1. Framing Business Questions

The project addressed questions such as:

Business Question	Visualization / Analysis
What is the overall level of employee attrition?	KPI cards showing total employees, employees who left and attrition rate.
How does attrition vary with salary growth?	Attrition Risk Across Salary Growth Bands.
How does attrition differ across education backgrounds?	Attrition Patterns Across Educational Backgrounds.
How does work mode relate to attrition and burnout?	Burnout Risk Across Different Work Models and Attrition Risk by Workplace Model.
How does attrition change with tenure and career level?	Employee Attrition Across Tenure and Attrition Trends Across Career Levels.
Does overtime relate to attrition?	Attrition Risk by Overtime.
Does performance level relate to attrition?	Attrition Risk Across Performance Level.
How does attrition vary across departments and roles?	Department Wise Attrition Rate and job-role analysis.

4.2. Developing Visualizations

Power BI was used to develop KPI cards, bar charts, line charts, a scatter-style educational analysis, a donut chart, slicers, and other analytical visuals. The visuals were arranged into dashboard pages so that users can move from overall workforce indicators to detailed attrition patterns.

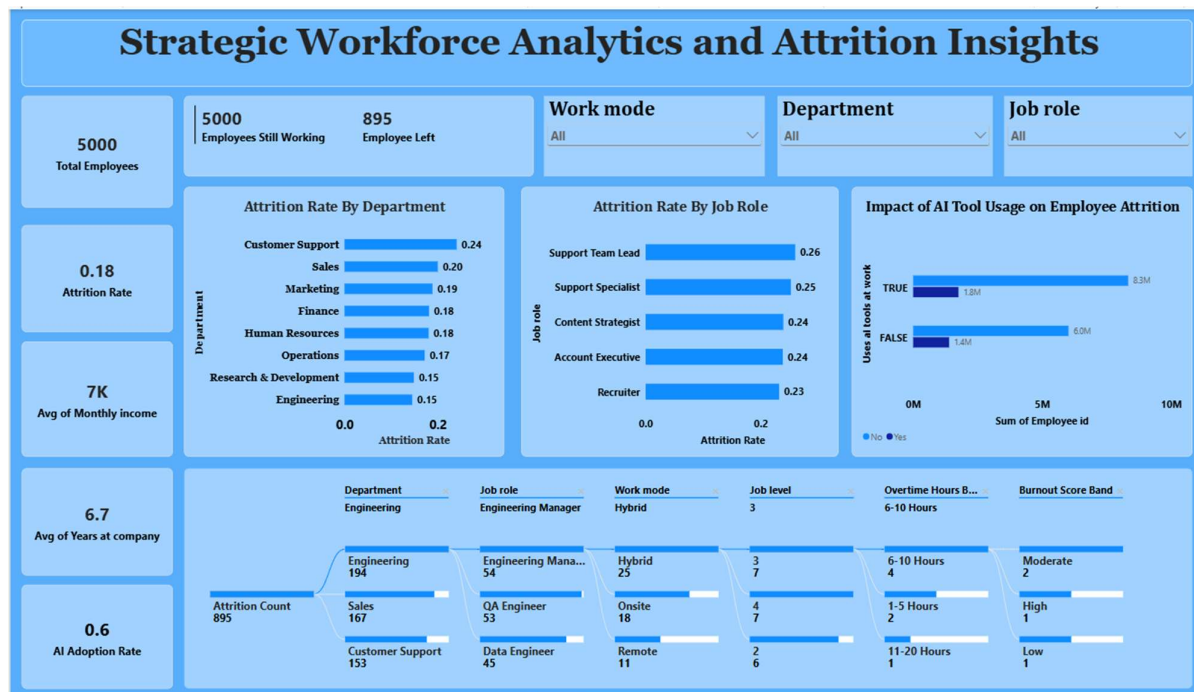
The first dashboard emphasizes employee retention performance and workforce wellbeing indicators. The second dashboard focuses on employee performance and career growth. The strategic workforce analysis page combines high-level workforce KPIs with department, job role and work-mode analysis.

5. Dashboard

5.1. Dashboard Design Screenshot with Insights

The below screenshot represents the developed Power BI dashboards.

- The dashboard shows total employees, employees who left and attrition rate.
- It analyzes attrition by department, job role and work mode. It also includes AI adoption analysis



Key dashboard observations include an overall attrition rate of approximately 18%, higher displayed attrition in lower salary-growth bands, elevated displayed attrition in the 21–30 hours overtime band, and differences across performance levels, work models, career levels and departments. These are descriptive associations shown by the dashboard.

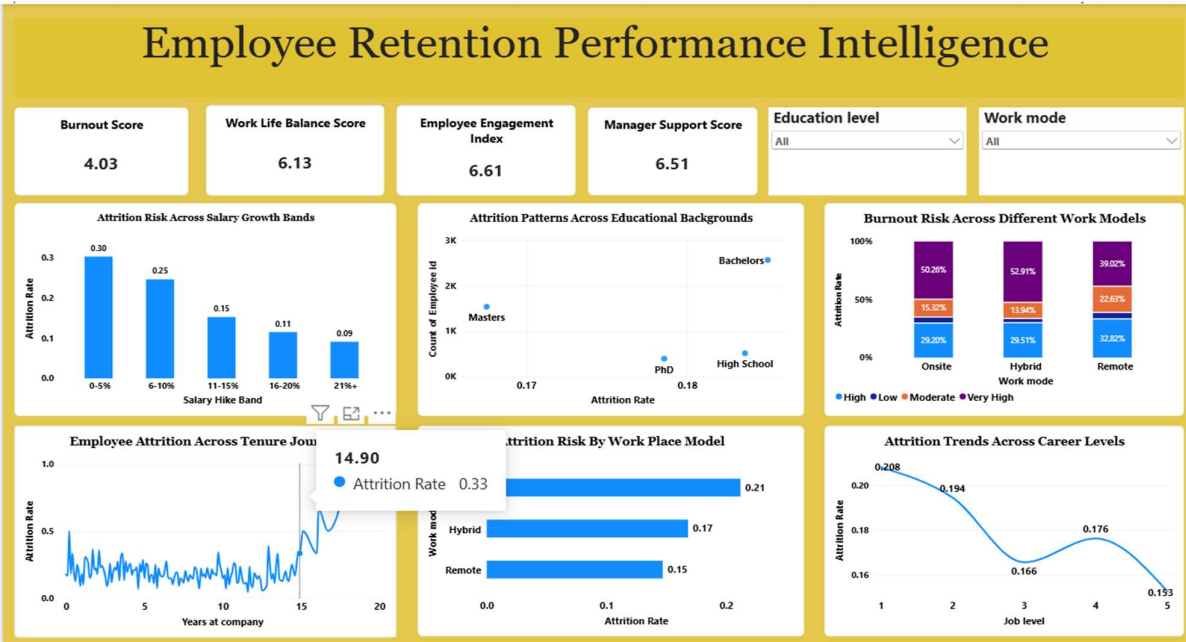
6. Report

6.1. Report Design Screenshot with Insights

Report 1 – Employee Retention Performance Intelligence

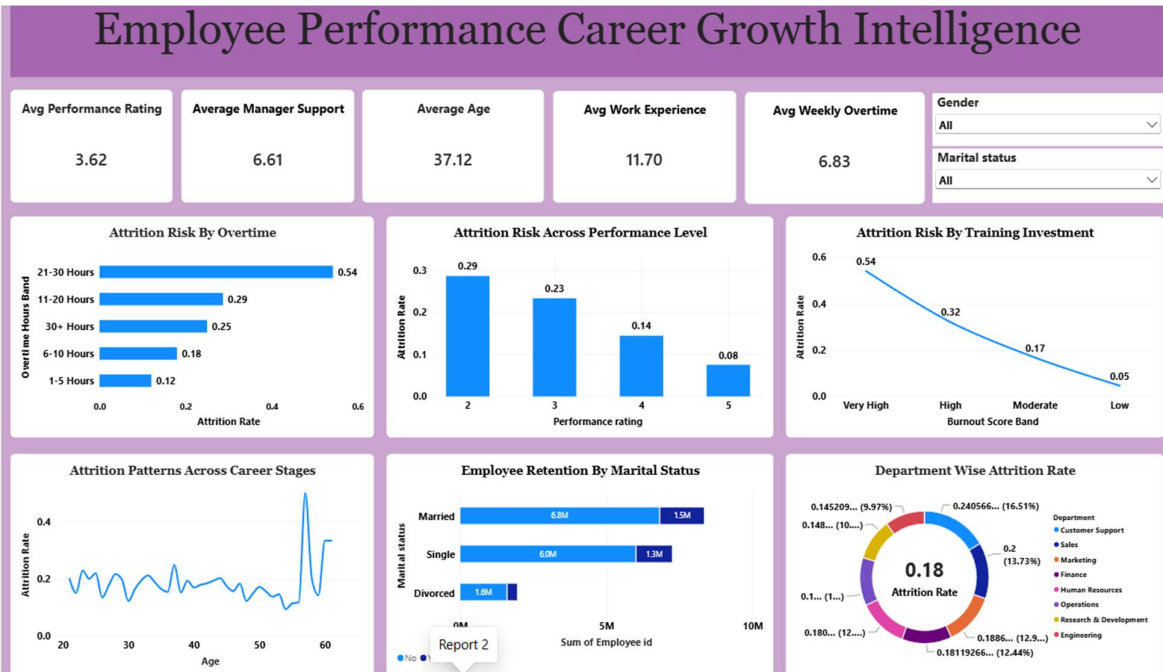
Analyzes burnout, engagement, work-life balance, manager support, salary growth, and work mode.

Helps identify factors associated with employee retention and attrition.



Report 2 – Employee Performance Career Growth Intelligence

Analyzes performance, manager support, experience, overtime, training, career stages, and department-wise attrition. Helps understand employee performance and career-related attrition patterns.



The report pages provide a detailed analytical view by combining related charts on the same page to support comparison between employee wellbeing, performance, career growth, workload, work mode and attrition.

The Employee Retention Performance Intelligence page highlights burnout score, work-life balance, employee engagement and manager support, followed by attrition analysis across salary growth, education, work model, tenure and career levels.

The Employee Performance Career Growth Intelligence page highlights performance rating, manager support, age, work experience and weekly overtime, followed by attrition analysis across overtime, performance level, training investment, career stages, marital status and department.

The Strategic Workforce Analysis and Attrition Insights page provides an overall workforce view with total employees, employees who left, attrition rate, average experience, average overtime and AI adoption rate, together with department, job role and work-mode analysis.

7. Performance Testing

7.1 Utilization of Data filters

Interactive filters were incorporated into the dashboards to allow users to analyze subsets of the workforce. The supplied dashboards include filters such as Education Level, Work Mode, Gender and Marital Status. These filters allow KPIs and visuals to be examined for selected employee segments.

7.2 No of Calculation Columns/Measures

The dashboard uses calculated KPI/aggregation logic for measures such as total employees, employees who left, attrition rate, average burnout score, work-life balance score, employee engagement, manager support, performance rating, work experience, weekly overtime and AI adoption rate. The exact number of calculation columns/measures is not visible in the supplied screenshots, so an exact count is not stated.

Employee Attrition Data	
☐ Σ Age	☐ Gender
☐ Σ AI Adoption Rate	☐ Σ Job level
☐ Attrition	☐ Job role
☐ Σ Attrition Count	☐ Σ Last review score
☐ Σ Attrition Rate	☐ Σ Manager support score
☐ Σ Burnout score	☐ Marital status
☐ Σ Burnout Score Band	☐ Σ Monthly income
☐ Σ Business travel days per year	☐ Σ Num prior companies
☐ Σ Commute minutes	☐ Σ Overtime Hours Band
☐ Department	☐ Σ Overtime hours per week
☐ Education level	☐ Σ Perceived ai job risk
☐ Σ Employee id	☐ Σ Performance rating
☐ Σ Employee Left	☐ Σ Salary hike (%)
☐ Σ Engagement score	☐ Σ Salary Hike Band
☐ Gender	☐ Σ Stock option level
☐ Σ Job level	☐ Σ Total working years
☐ Job role	☐ Σ Training hours
☐ Σ Last review score	☐ Uses ai tools at work
☐ Σ Manager support score	☐ Σ Work life balance score
☐ Marital status	☐ Work mode
☐ Σ Monthly income	☐ Σ Years at company
☐ Σ Num prior companies	☐ Σ Years since promotion

7.3 No of Visualization

The final solution uses multiple KPI cards, slicers, bar charts, line charts, a donut chart and other analytical visuals across the dashboard pages. The exact total number of visuals cannot be reliably counted from the supplied screenshots alone; the final Power BI file should be used for the exact count.

8. Conclusion/Observation

The Strategic Employee Attrition Intelligence Dashboard provides a consolidated view of employee retention and workforce performance. The analysis shows that attrition can be examined across several dimensions rather than through a single overall percentage.

- The workforce contains 5,000 employees, with 895 employees reported as having left, corresponding to an overall attrition rate of 17.9% in the project problem statement.
- The dashboard shows differences in attrition across salary-growth bands, with the lowest salary-growth band showing the highest displayed attrition rate.
- Overtime is associated with different displayed attrition levels, with the 21–30 hours overtime band showing a displayed attrition rate of 0.54.
- Attrition varies across performance ratings, with the displayed rate decreasing from performance rating 2 toward rating 5.
- The dashboards show differences in attrition across onsite, hybrid and remote work models.
- Attrition also varies across career levels, tenure and departments, providing multiple areas for HR segmentation and further investigation.
- The dashboard provides AI adoption as one of the workforce indicators, allowing technology adoption to be considered alongside broader employee and workplace measures.

These observations are descriptive findings from the dashboard. They identify patterns and areas for further investigation; they do not by themselves establish that any single factor causes employee attrition.

9. Future Scope

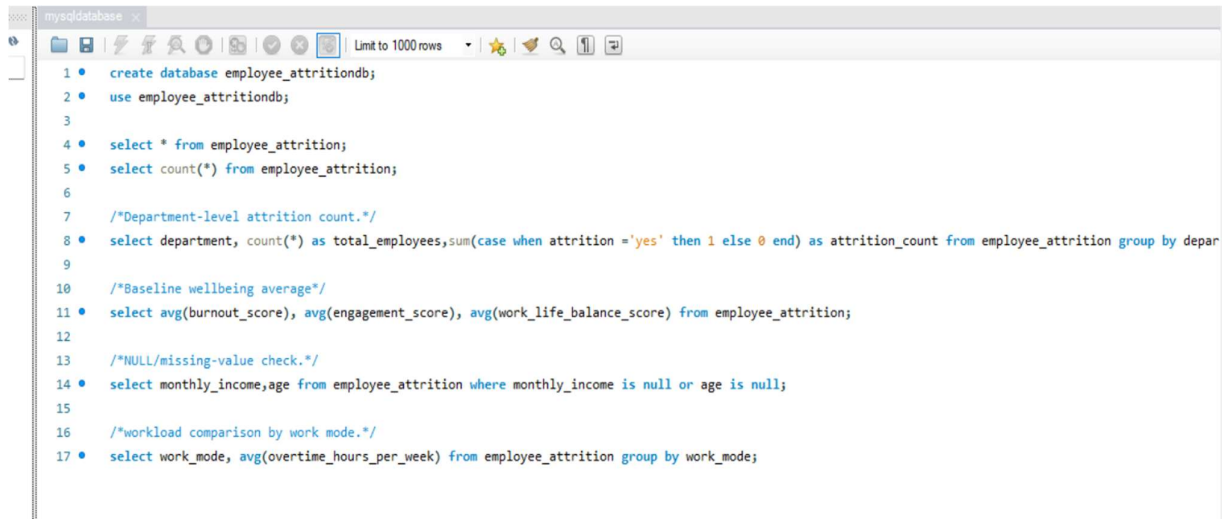
- Develop a predictive attrition model to estimate individual or segment-level attrition risk using historical employee data.
- Add time-based workforce tracking to monitor whether attrition is increasing or decreasing over future periods.
- Integrate additional HR sources such as employee surveys, compensation history, promotion history, absenteeism and exit-interview information.
- Enhance AI-assisted analysis to automatically summarize major changes, unusual segments and potential retention-risk areas.
- Introduce role-based access so HR managers and leadership can view information relevant to their responsibilities.
- Add automated refresh and deployment so that the dashboard can be maintained as new employee data becomes available.

10. Appendix

10.1. Source Code (if any)

SQL: Used for data analysis and data exploration.

SQL quires

A screenshot of a MySQL database management tool interface. The window title is 'mysqldb'. The toolbar shows various icons for file operations, editing, and viewing. A dropdown menu indicates 'Limit to 1000 rows'. The main area displays a list of 17 SQL queries, numbered 1 through 17. The queries include database creation, table usage, data selection, aggregation, and conditional logic.

```
1 • create database employee_attritiondb;
2 • use employee_attritiondb;
3
4 • select * from employee_attrition;
5 • select count(*) from employee_attrition;
6
7 /*Department-level attrition count.*/
8 • select department, count(*) as total_employees, sum(case when attrition = 'yes' then 1 else 0 end) as attrition_count from employee_attrition group by departmen
9
10 /*Baseline wellbeing average*/
11 • select avg(burnout_score), avg(engagement_score), avg(work_life_balance_score) from employee_attrition;
12
13 /*NULL/missing-value check.*/
14 • select monthly_income, age from employee_attrition where monthly_income is null or age is null;
15
16 /*workload comparison by work mode.*/
17 • select work_mode, avg(overtime_hours_per_week) from employee_attrition group by work_mode;
```

Power BI:

The project was developed primarily using Power BI for data preparation, calculations, visualization and dashboard/report development. No separate application source code was provided in the supplied final-report materials. Any Power Query transformations, DAX measures, calculated columns or other Power BI expressions can be included here from the final Power BI project file if required.

10.2. GitHub & Project Demo Link

GitHub Repository:

<https://github.com/chakka-priyanka/Strategic-Employee-Attrition-Intelligence-Dashbord>

Project Demo Link: